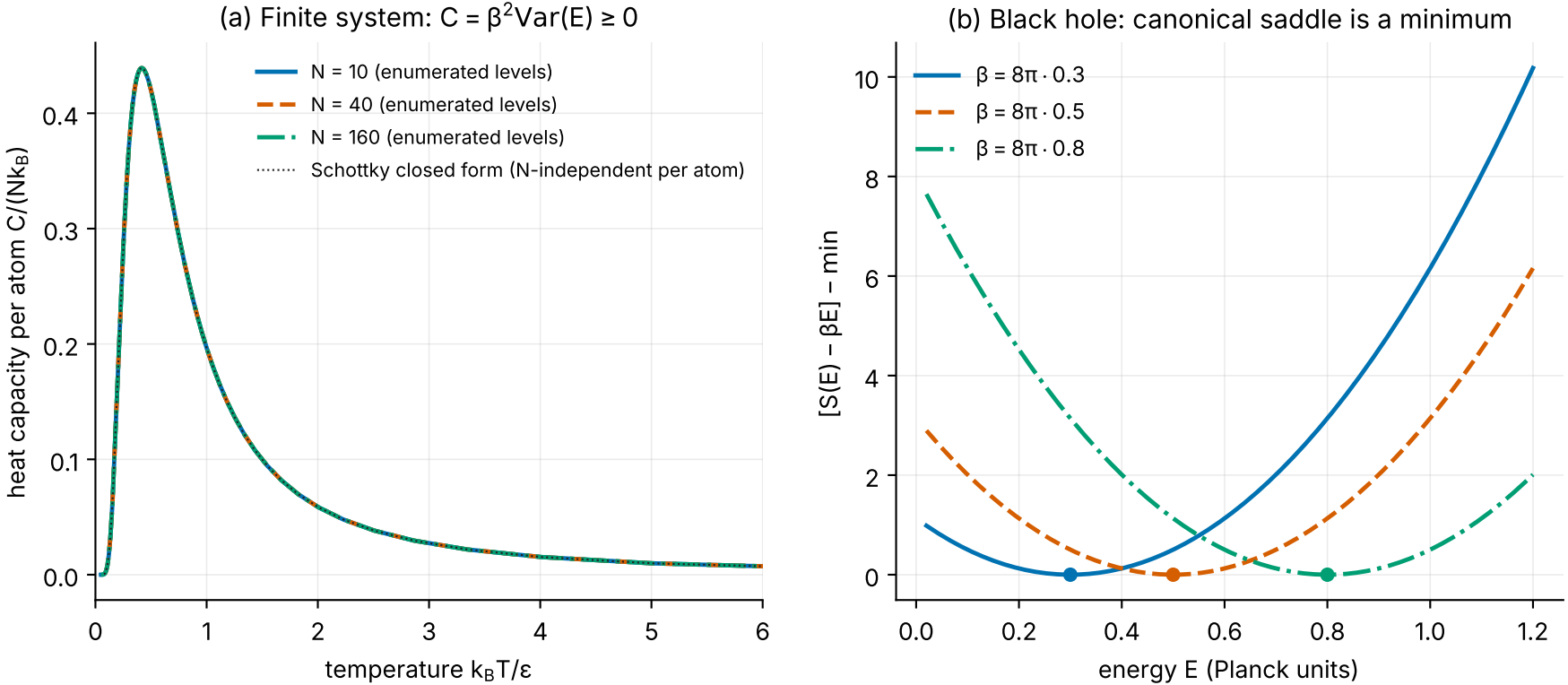
(a) Finite system: C = β²Var(E) ≥ 0
N = 10 (enumerated levels)
N = 40 (enumerated levels)
N = 160 (enumerated levels)
Schottky closed form (N-independent per atom)
heat capacity per atom C/(Nk_B)
temperature k_BT/ε
0.0
0.1
0.2
0.3
0.4
0
1
2
3
4
5
6
(b) Black hole: canonical saddle is a minimum
β = 8π · 0.3
β = 8π · 0.5
β = 8π · 0.8
[S(E) − βE] − min
energy E (Planck units)
0
2
4
6
8
10
0.0
0.2
0.4
0.6
0.8
1.0
1.2